R Notebook

Dhruvi Sheth 22294

```r
getwd()
```

```
## [1] "/home/dhruvi/Desktop/Biostatistics/GapMinder"
```

```r
lifeExpectancy <- read.csv("life_expectancy_years.csv", header = TRUE, check.names = FALSE)
```

```r
lifeExpectancy[lifeExpectancy$country == "Sweden"]
```

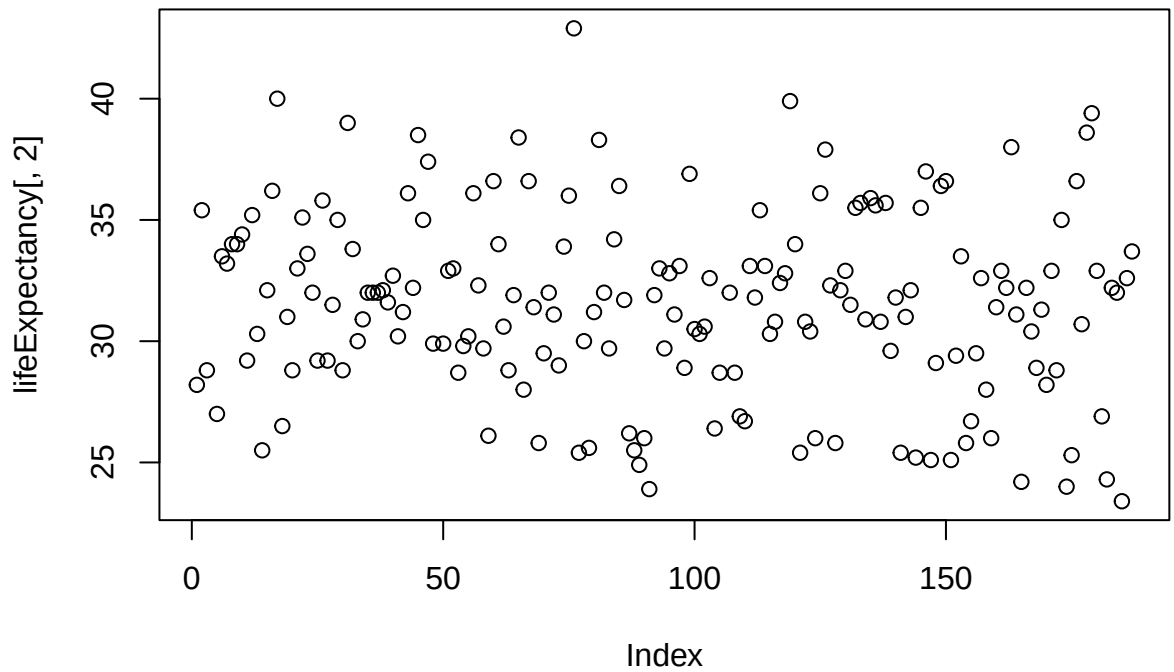
```
##     1960
## 1   39.3
## 2   62.2
## 3   52.5
## 4     NA
## 5   40.6
## 6   63.3
## 7   64.2
## 8   61.9
## 9   71.1
## 10  69.0
## 11  58.7
## 12  62.4
## 13  53.3
## 14  43.9
## 15  64.7
## 16  71.0
## 17  69.7
## 18  61.0
## 19  40.9
## 20  41.7
## 21  46.0
## 22  61.7
## 23  51.5
## 24  56.1
## 25  58.8
## 26  69.2
## 27  38.7
## 28  42.6
## 29  44.1
## 30  45.5
## 31  71.0
## 32  55.4
## 33  40.6
## 34  44.6
```

```
## 35  57.9
## 36  31.6
## 37  61.0
## 38  45.9
## 39  46.6
## 40  44.7
## 41  63.3
## 42  42.3
## 43  64.6
## 44  65.6
## 45  68.5
## 46  70.3
## 47  72.0
## 48  51.7
## 49    NA
## 50  57.8
## 51  58.0
## 52  46.9
## 53  53.9
## 54  40.5
## 55  39.1
## 56  69.1
## 57  46.9
## 58  38.4
## 59  61.7
## 60  68.8
## 61  70.6
## 62  44.1
## 63  44.0
## 64  61.3
## 65  69.4
## 66  51.1
## 67  70.8
## 68  61.9
## 69  43.6
## 70  42.1
## 71  38.6
## 72  60.4
## 73  41.2
## 74  51.2
## 75  68.0
## 76  74.0
## 77  41.9
## 78  46.0
## 79  48.6
## 80  49.6
## 81  70.4
## 82  68.6
## 83  69.2
## 84  65.2
## 85  68.3
## 86  53.7
## 87  60.0
## 88  51.3
```

```
## 89  50.0
## 90  62.6
## 91  55.7
## 92  39.4
## 93  70.5
## 94  64.7
## 95  52.4
## 96  40.6
## 97  48.8
## 98  70.3
## 99  69.7
## 100 44.7
## 101 40.2
## 102 59.4
## 103 33.5
## 104 34.0
## 105 67.3
## 106   NA
## 107 47.3
## 108 57.3
## 109 57.6
## 110 56.2
## 111 60.5
## 112 46.8
## 113 65.5
## 114 50.3
## 115 41.4
## 116 38.8
## 117 51.6
## 118 44.2
## 119 73.5
## 120 71.2
## 121 49.7
## 122 41.5
## 123 49.6
## 124 57.4
## 125 62.6
## 126 73.4
## 127 41.9
## 128 48.6
## 129 54.4
## 130 63.8
## 131 43.3
## 132 67.2
## 133 43.3
## 134 60.3
## 135 68.0
## 136 64.3
## 137 57.8
## 138 65.9
## 139 68.7
## 140 44.4
## 141 59.5
## 142 52.5
```

```
## 143 54.6
## 144 46.2
## 145 63.4
## 146 61.8
## 147 38.7
## 148 66.5
## 149 70.3
## 150 67.7
## 151 49.5
## 152 43.7
## 153 53.3
## 154 53.8
## 155 42.0
## 156 68.9
## 157 58.6
## 158 58.5
## 159 58.0
## 160 48.5
## 161 59.7
## 162 73.0
## 163 71.6
## 164 56.4
## 165 54.4
## 166 48.4
## 167 57.7
## 168 38.4
## 169 44.6
## 170 63.8
## 171 63.6
## 172 47.2
## 173 48.1
## 174 54.0
## 175 43.9
## 176 69.8
## 177 53.9
## 178 71.0
## 179 70.0
## 180 67.9
## 181 60.2
## 182 50.5
## 183 60.7
## 184 54.4
## 185 33.0
## 186 49.2
## 187 53.2
```

```r
plot(lifeExpectancy[,2])
```

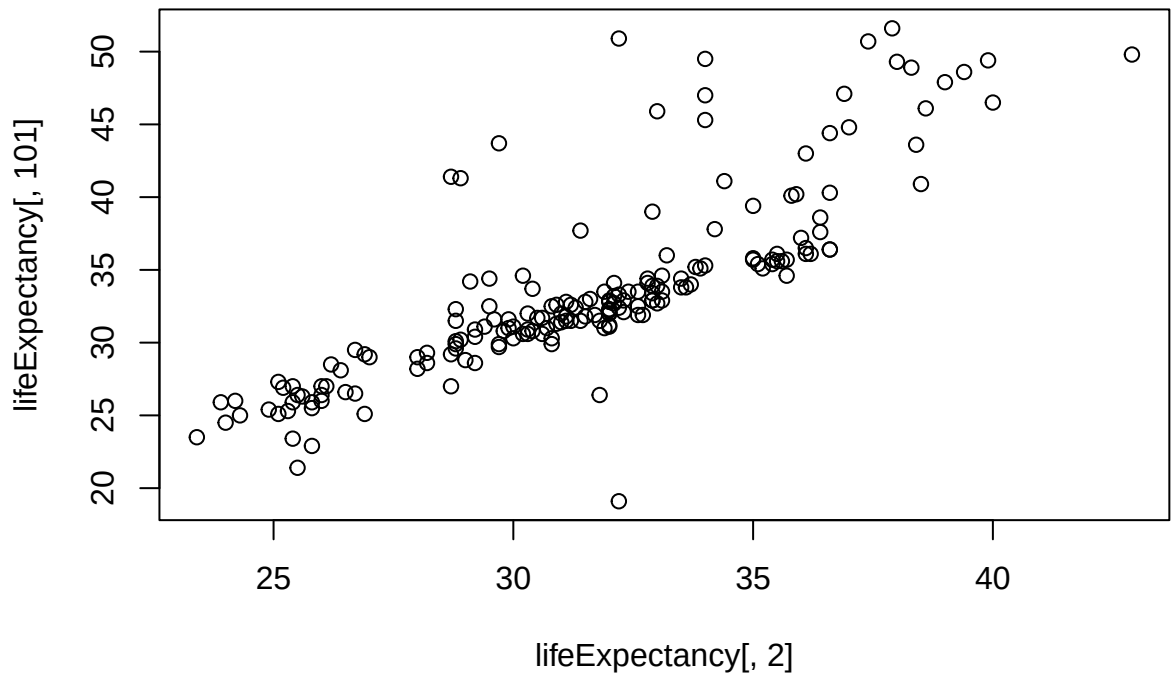



```
years <- as.numeric(colnames(lifeExpectancy)[2:302])
le <- as.numeric(lifeExpectancy[162, 2:302])

plot(as.vector(unlist(lifeExpectancy[162,2:302])),
     as.vector(unlist(colnames(lifeExpectancy)[2:302]))
     ,xlab="Life expectancy in years", ylab="Year")
abline(h=2022)
```

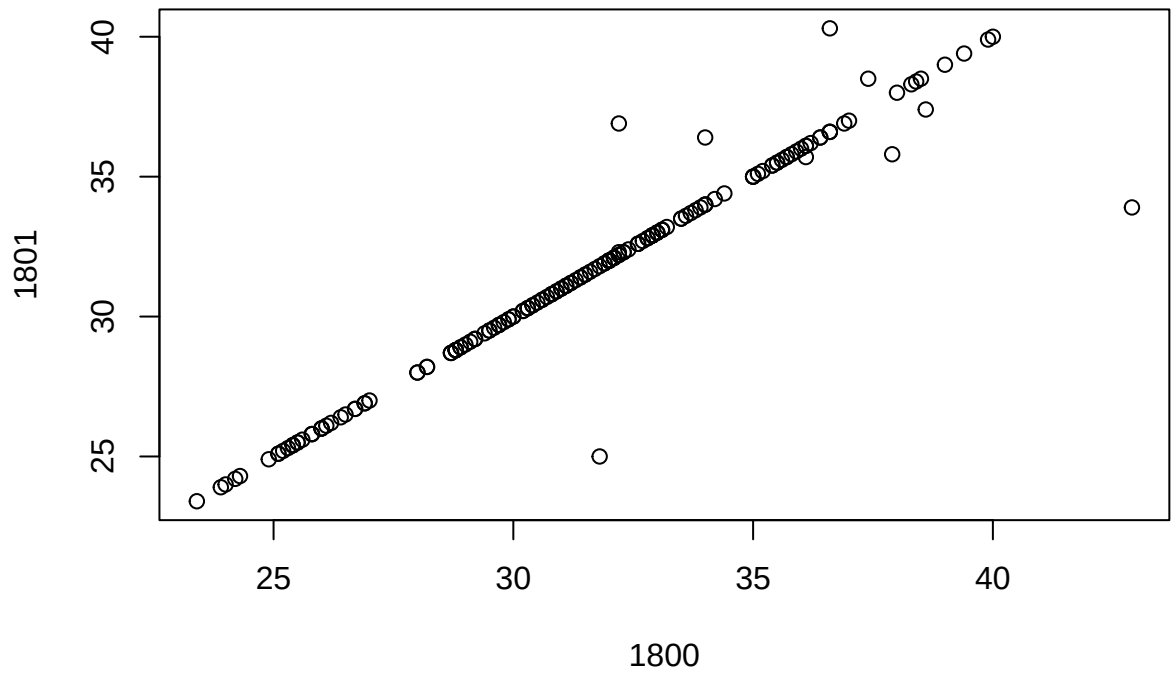
```
lifeExpectancy[162,"2020"]
```

```
## [1] 82.9
```

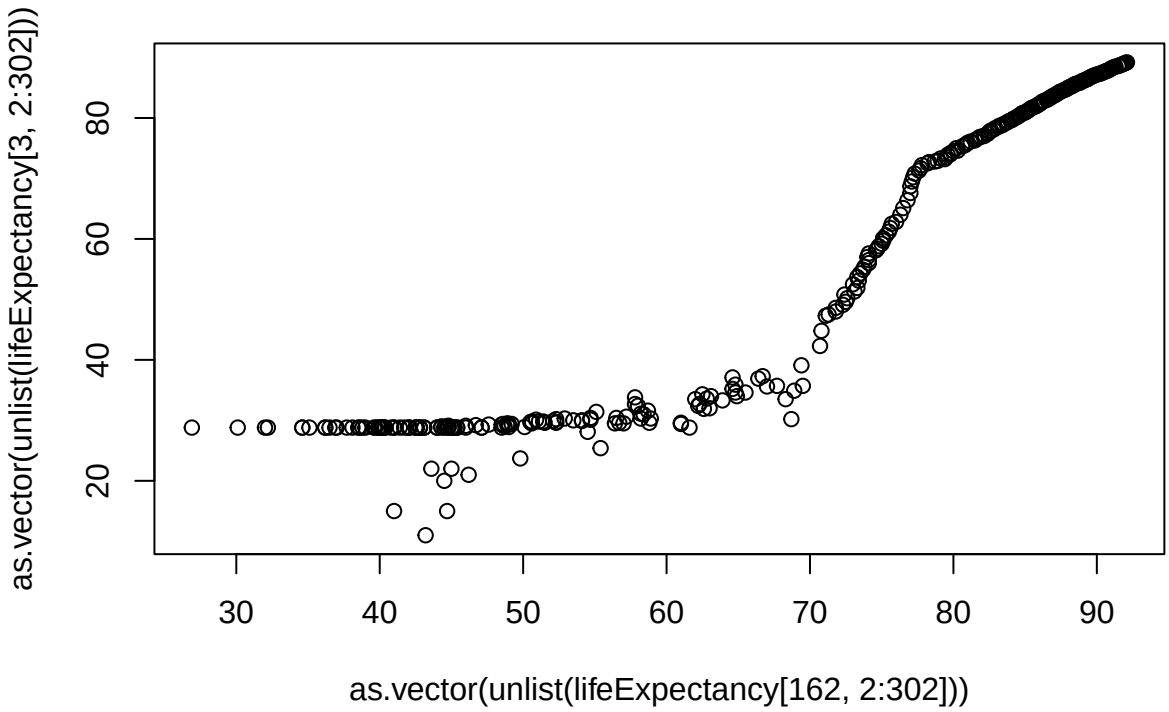
```
plot(lifeExpectancy[,2],lifeExpectancy[,101])
```



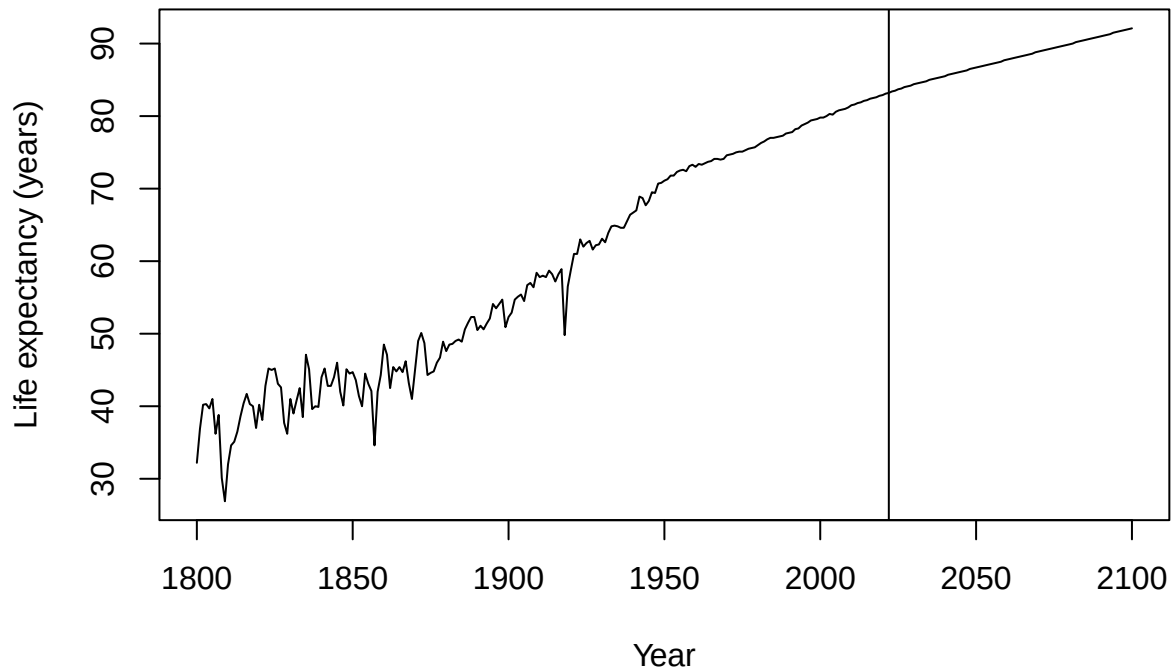

```
plot(lifeExpectancy[,2],lifeExpectancy[,3],xlab="1800", ylab="1801")
```




```
plot(as.vector(unlist(lifeExpectancy[162,2:302])),as.vector(unlist(lifeExpectancy[3,2:302])))
```




```r
years <- as.numeric(colnames(lifeExpectancy)[2:302])
le <- as.numeric(lifeExpectancy[162, 2:302])

plot(years, le,
     xlab = "Year",
     ylab = "Life expectancy (years)",
     type = "l")

abline(v = 2022)
```




```
lifeExpectancy[lifeExpectancy$country=="Sweden",]
```

```
##     country 1800 1801 1802 1803 1804 1805 1806 1807 1808 1809 1810 1811 1812
## 162  Sweden 32.2 36.9 40.2 40.3 39.7   41 36.2 38.8 30.1 26.9   32 34.6 35.1
##     1813 1814 1815 1816 1817 1818 1819 1820 1821 1822 1823 1824 1825 1826 1827
## 162 36.5 38.6 40.4 41.7 40.3   40   37 40.2 38.1 42.8 45.2   45 45.2 43.1 42.6
##     1828 1829 1830 1831 1832 1833 1834 1835 1836 1837 1838 1839 1840 1841 1842
## 162 37.7 36.2   41   39 40.8 42.5 38.5 47.1 45.1 39.6   40 39.9   44 45.2 42.8
##     1843 1844 1845 1846 1847 1848 1849 1850 1851 1852 1853 1854 1855 1856 1857
## 162 42.8   44   46   42 40.1 45.1 44.5 44.7 43.6 41.4   40 44.5 43.1 42.1 34.6
##     1858 1859 1860 1861 1862 1863 1864 1865 1866 1867 1868 1869 1870 1871 1872
## 162   42 44.3 48.5 47.1 42.5 45.4 44.8 45.4 44.7 46.2 43.2   41   45   49 50.1
##     1873 1874 1875 1876 1877 1878 1879 1880 1881 1882 1883 1884 1885 1886 1887
## 162 48.7 44.3 44.6 44.8   46 46.7 48.9 47.6 48.5 48.6   49 49.2 48.9 50.6 51.5
##     1888 1889 1890 1891 1892 1893 1894 1895 1896 1897 1898 1899 1900 1901 1902
## 162 52.3 52.3 50.5 51.1 50.6 51.4 52.1 54.1 53.5 54.1 54.7 50.9 52.3 52.9 54.7
##     1903 1904 1905 1906 1907 1908 1909 1910 1911 1912 1913 1914 1915 1916 1917
## 162 55.1 55.4 54.5 56.7   57 56.4 58.4 57.8   58 57.8 58.7 58.2 57.2 58.2 58.9
##     1918 1919 1920 1921 1922 1923 1924 1925 1926 1927 1928 1929 1930 1931 1932
## 162 49.8 56.5 58.8   61   61   63   62 62.5 62.8 61.6 62.2 62.3 63.1 62.6 63.9
##     1933 1934 1935 1936 1937 1938 1939 1940 1941 1942 1943 1944 1945 1946 1947
## 162 64.8 64.9 64.8 64.6 64.6 65.5 66.4 66.7   67 68.9 68.7 67.7 68.3 69.5 69.4
##     1948 1949 1950 1951 1952 1953 1954 1955 1956 1957 1958 1959 1960 1961 1962
## 162 70.7 70.8 71.1 71.3 71.8 71.8 72.3 72.5 72.6 72.4 73.1 73.3   73 73.4 73.3
##     1963 1964 1965 1966 1967 1968 1969 1970 1971 1972 1973 1974 1975 1976 1977
```

```
## 162 73.5 73.7 73.8 74.1 74.1   74 74.1 74.6 74.7 74.8   75 75.1 75.1 75.3 75.5
##     1978 1979 1980 1981 1982 1983 1984 1985 1986 1987 1988 1989 1990 1991 1992
## 162 75.6 75.7   76 76.3 76.5 76.8   77   77 77.1 77.2 77.3 77.6 77.7 77.8 78.2
##     1993 1994 1995 1996 1997 1998 1999 2000 2001 2002 2003 2004 2005 2006 2007
## 162 78.3 78.7 78.9 79.1 79.4 79.5 79.6 79.8 79.8   80 80.3 80.2 80.6 80.8 80.9
##     2008 2009 2010 2011 2012 2013 2014 2015 2016 2017 2018 2019 2020 2021 2022
## 162   81 81.2 81.5 81.6 81.8 81.9 82.1 82.2 82.4 82.5 82.6 82.8 82.9 83.1 83.2
##     2023 2024 2025 2026 2027 2028 2029 2030 2031 2032 2033 2034 2035 2036 2037
## 162 83.4 83.5 83.7 83.8   84 84.1 84.2 84.4 84.5 84.6 84.7 84.8   85 85.1 85.2
##     2038 2039 2040 2041 2042 2043 2044 2045 2046 2047 2048 2049 2050 2051 2052
## 162 85.3 85.4 85.5 85.7 85.8 85.9   86 86.1 86.2 86.3 86.5 86.6 86.7 86.8 86.9
##     2053 2054 2055 2056 2057 2058 2059 2060 2061 2062 2063 2064 2065 2066 2067
## 162   87 87.1 87.2 87.3 87.4 87.5 87.7 87.8 87.9   88 88.1 88.2 88.3 88.4 88.5
##     2068 2069 2070 2071 2072 2073 2074 2075 2076 2077 2078 2079 2080 2081 2082
## 162 88.6 88.8 88.9   89 89.1 89.2 89.3 89.4 89.5 89.6 89.7 89.8 89.9   90 90.2
##     2083 2084 2085 2086 2087 2088 2089 2090 2091 2092 2093 2094 2095 2096 2097
## 162 90.3 90.4 90.5 90.6 90.7 90.8 90.9   91 91.1 91.2 91.3 91.5 91.6 91.7 91.8
##     2098 2099 2100
## 162 91.9   92 92.1
```

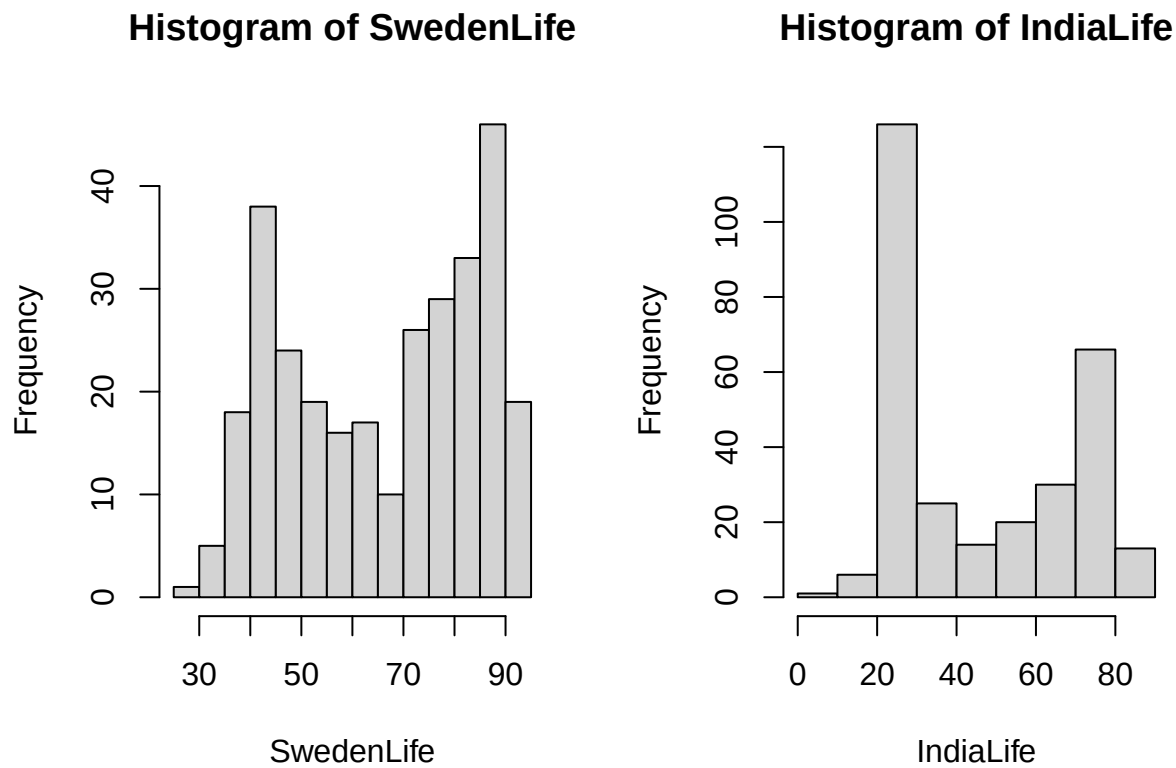
```r
na.omit(as.numeric(unlist(lifeExpectancy[lifeExpectancy$country=="Sweden",])))->SwedenLife
```

```
## Warning in na.omit(as.numeric(unlist(lifeExpectancy[lifeExpectancy$country == :
## NAs introduced by coercion
```

```r
na.omit(as.numeric(unlist(lifeExpectancy[lifeExpectancy$country=="India",])))->IndiaLife
```

```
## Warning in na.omit(as.numeric(unlist(lifeExpectancy[lifeExpectancy$country == :
## NAs introduced by coercion
```

```r
par(mfrow=c(1,2))
hist(SwedenLife)
hist(IndiaLife)
```




```
na.omit(as.numeric(unlist(lifeExpectancy[lifeExpectancy$country=="Bangladesh",])))->BanglaLife
```

```
## Warning in na.omit(as.numeric(unlist(lifeExpectancy[lifeExpectancy$country == :
## NAs introduced by coercion
```

```
na.omit(as.numeric(unlist(lifeExpectancy[lifeExpectancy$country=="Australia",])))->AussieLife
```

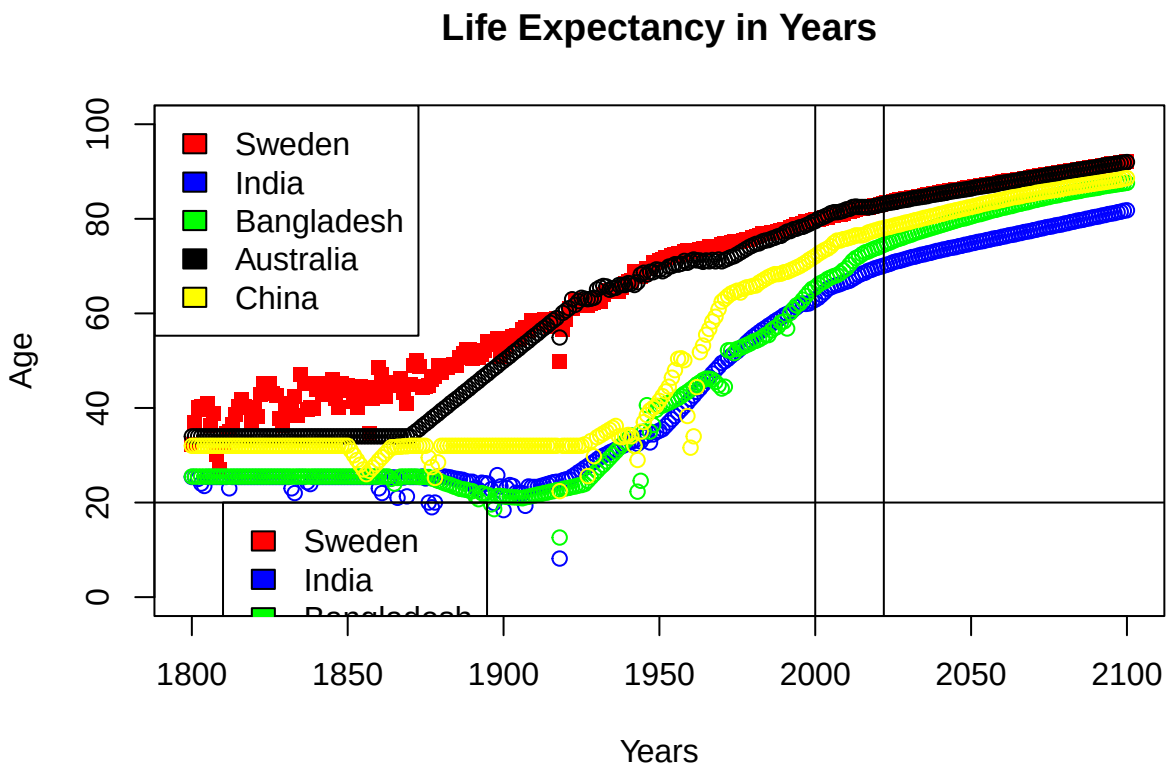
```
## Warning in na.omit(as.numeric(unlist(lifeExpectancy[lifeExpectancy$country == :
## NAs introduced by coercion
```

```
na.omit(as.numeric(unlist(lifeExpectancy[lifeExpectancy$country=="China",])))->ChinaLife
```

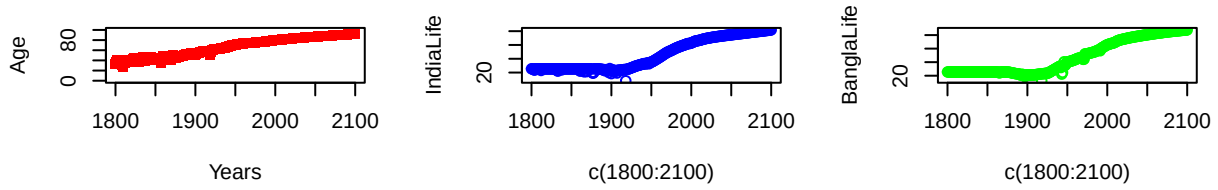
```
## Warning in na.omit(as.numeric(unlist(lifeExpectancy[lifeExpectancy$country == :
## NAs introduced by coercion
```

```
plot(c(1800:2100),SwedenLife,col="red",main="Life Expectancy in Years",pch=15,ylim=c(0,100),xlab="Years
points(c(1800:2100),IndiaLife,col="blue")
points(c(1800:2100),BanglaLife,col="green")
points(c(1800:2100),AussieLife,col="black")
points(c(1800:2100),ChinaLife,col="yellow")

legend(x = "topleft",          # Position
       legend = c("Sweden", "India", "Bangladesh", "Australia", "China"),  # Legend texts
```

```
       fill = c("red", "blue","green", "black", "yellow"))              # Colors

legend(x =1810, y=20,          # Position
       legend = c("Sweden", "India", "Bangladesh", "Australia", "China"),  # Legend texts
       fill = c("red", "blue","green", "black", "yellow"))              # Colors


abline(v=2000)
abline(v=2022)

abline(h=20)
```




```
par(mfrow=c(3,3))
plot(c(1800:2100),SwedenLife,col="red",pch=15,ylim=c(0,100),xlab="Years",ylab="Age")
plot(c(1800:2100),IndiaLife,col="blue")
plot(c(1800:2100),BanglaLife,col="green")
```


Age
0
80
1800
1900
2000
2100
Years
IndiaLife
20
1800
1900
2000
2100
c(1800:2100)
BanglaLife
20
1800
1900
2000
2100
c(1800:2100)